

Critical Reflection Report: Acting as Creative Director in AI-Assisted Product Development

Overview

This reflection examines the experience of building a digital product through AI-assisted prototyping while taking on the role of **creative director** rather than manual coder. In this workflow, the main responsibility shifts from writing each line of code to defining the problem clearly, setting constraints, shaping UX direction, validating outputs, and correcting weak or misleading generations. Research on AI prototyping suggests that these tools are strongest in ideation and early concept exploration, but they still require human judgment to resolve nuance, quality, and usability gaps.

The Decision Debt Calculator project illustrates this shift well. The concept itself is well suited to AI prototyping because it has a clear user problem, structured inputs, deterministic scoring logic, and visible outputs such as rankings, charts, and recommendation text. That combination makes it easy for a generative tool to produce a functioning first version quickly, while still leaving important decisions about structure, interaction quality, and trustworthiness in human hands.

Roadblocks and Hallucinations

The first major roadblock in an AI-assisted workflow is that the model often produces something that looks complete before it is actually correct. In decision-support products, this can appear as broken weighted scoring, inconsistent scenario presets, missing edge-case handling, or recommendation text that sounds persuasive but does not truly match the underlying logic. Hallucinations in this context are not only factual inventions; they also include fabricated UX confidence, where the interface implies rigor even when the calculations or state transitions are flawed.

A second roadblock is design conventionality. AI systems tend to default toward mainstream, repetitive UI patterns, which can lead to generic dashboard layouts, weak hierarchy, and surface-level polish without a distinct product point of view. This creates a gap between “working prototype” and “convincing product,” especially for tools meant to support serious decisions such as career moves, business choices, or education planning.

Another recurring issue is that the model may satisfy the wording of the prompt while missing the intent behind it. For example, asking for a recommendation engine can result in static summary text instead of a dynamic explanation tied to actual score changes. Asking for mobile responsiveness may produce a nominally responsive layout that still becomes unusable when a scoring matrix grows wide or criteria are added. These issues show that AI follows instructions best when requirements are decomposed into explicit behavioural rules, not just feature labels.

Prompting Strategy Adjustments

To overcome these failures, the prompting strategy has to evolve from broad requests into layered direction. A generic request such as “build a career decision app” is too under-specified; it invites the model to fill in gaps with assumptions. A stronger prompt defines user problem, app type, workflow steps, scoring model, required UI sections, logic functions, edge cases, and design constraints. This reduces ambiguity and gives the model a narrower space in which to improvise.

Another effective adjustment is separating product intent from implementation detail. Instead of prompting only for screens or only for code, the better approach is to state what the product must help the user accomplish, how the system should reason, and how success should be evaluated. Research on AI prototyping emphasizes that high-quality outputs depend heavily on prior thinking, such as drafting a lightweight PRD, clarifying assumptions, and specifying interactions before generation begins.

Prompting also becomes more iterative and diagnostic. When the first output has broken logic or awkward UX, the correction prompt works best when it identifies the exact failure mode: for

example, “rebuild the scoring engine so all rankings derive from weighted criteria only,” “make scenario toggles rebalance weights while preserving manual edits,” or “rewrite explanation text so it references the top strengths and trade-offs of the current winner.” This is consistent with guidance on reducing hallucinations through boundaries, explicit source anchoring, iterative testing, and validation-focused instructions.

A final adjustment is treating prompting as direction-setting rather than one-time generation. The role becomes similar to a product lead or creative director: define the standard, review the prototype critically, identify what feels untrustworthy or generic, and then steer the next iteration with sharper constraints. In this model, prompting is less about asking for output and more about continuously shaping judgment.

Perspective on Product Development

This workflow changes the understanding of digital product development in a meaningful way. Traditional development often foregrounds execution effort: wireframing, coding, styling, debugging, and iteration are treated as the main bottlenecks. AI-assisted prototyping lowers the cost of producing a first working version, which shifts value toward problem framing, system thinking, prioritization, and quality control.

It also makes rapid prototyping more strategic. Because a prototype can be assembled quickly, the real question becomes not “Can this be built?” but “What exactly should be built, for whom, and under what assumptions?” This pushes the builder to think earlier about user trust, edge cases, decision transparency, and whether the experience genuinely helps someone make a better choice rather than simply presenting attractive UI.

At the same time, the workflow does not eliminate the need for expertise. In fact, it can amplify the advantage of someone who already understands UX, product logic, information hierarchy, and evaluation criteria. AI can accelerate output, but it still depends on human ability to judge whether the output is coherent, ethical, usable, and aligned with the product’s purpose.

Implications for Future Practice

The strongest lesson from this process is that AI-assisted building works best when treated as a collaboration with strict supervision. Early generations are useful for momentum, ideation, and structure, but they should not be mistaken for finished solutions. For serious products, especially those involving recommendations or decision support, every generated interaction and explanation needs human review because confident output can still mask shallow reasoning.

This suggests a future workflow in which product builders maintain reusable prompt frameworks, decision rules, and evaluation checklists alongside prototypes. Instead of starting from a blank page each time, they build a system of constraints that captures product intent, design taste, and technical guardrails. That approach makes each new prototype faster to produce and more reliable to refine. In practical terms, acting as a creative director in this process means owning the standards rather than the syntax. The critical work lies in identifying what matters, articulating it clearly, and pushing the prototype toward usefulness and trust. That shift does not reduce the importance of craftsmanship; it relocates craftsmanship from manual production alone to direction, discernment, and iteration.